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NE 763 Exam 4

Please provide your responses to the following questions. Point values indicated the expected depth of response.

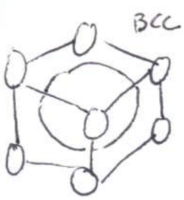
1. What are some considerations when optimizing the composition for F/M steels? (10 pts) - Ni, Si, Cr
2. Why do ferritic steels swell considerably less than austenitic steels? (10 pts)
3. What is the ductile to brittle transition temperature? Discuss irradiation embrittlement in F/M alloys. (10 pts)
4. Why are FeCrAl alloys of interest? What are the considerations for balancing compositions of these alloys? (8 pts)
5. What role do the oxide particles play in ODS steels? (6 pts)
6. What are some advantages and disadvantages of Ni alloys? (8 pts) - precipitates dissolve at ↑T - low T - reform hardness ↓ w/T?
7. What are benefits and drawbacks of U-Si fuel compared to U-Al fuel? Describe the differences in U3Si and U3Si2 swelling. (12 pts)
8. The gamma phase of U-Mo is not the thermodynamically stable phase at research reactor temperatures. Why is this phase the dominant phase in-reactor? (7 pts)
9. What effect does the solidus/liquidus gap have on the fabrication of U-Mo fuels? (6 pts)
10. Discuss the evolution of fission gas bubbles in U-Mo fuel as a function of burnup. (10 pts)
11. Why is the interaction layer of interest in U-Mo fuels? What mitigation strategies have been deployed to limit the amount of interaction layer? (7 pts)
12. Why is Al ideally suited for the research reactor environment when it is unable to be used in LWRs? (6 pts)

9/10

① Both heat treatments and compositional changes can be made to change the microstructure of F/M steels. Ideally, you want to add materials that help stabilize the ferrite phase ~~while~~ also maintaining the martensite laths. Basically you want the stability in operating conditions of the ferrite phase why also maintaing the strength of martensite (which has a high dislocation density). There are many elements that can be added to alter the microstructure and change properties such as oxidation resistance, creep resistance, etc. F/m steels should also be cheap ~~so cheaper to~~ to manufacture. - precipitation + strengthening

7/10

② Ferritic steels have a BCC structure which have much better resistance to swelling than FCC - austenitic structures...



BCC decreases dislocation movement and less migration and formation of voids. Whereas FCC has more dislocation move (Less packed structure), thus more ~~voids can~~ voids can form and grow, leading to swelling. - not enough

3) DBTT is ~~where~~ the temperature at which you transition from a ~~a~~ ductile material ~~material~~ to a brittle material where fracture occurs. This ~~[illegible]~~ is altered less by irradiation in F/M steels but can increase in other steels. DBTT is ~500-700 C for ferrite steels.

5/10 -needed much more

4) FeCrAl alloys are much less prone to oxidation than Zircaloy when exposed to high temperature steam environments (like those experienced during a LOCA). The Cr and Al help stabilize the ferrite phase which is less prone to swelling. Additionally the Al reacts with steam to for a protective Al_2O_3 layer on the surface of the cladding to slow oxidation and decrease an oxidation reaction. However too much aluminum can result in Al intermetallics within the structure, and to much Cr can accelerate embrittlement.

8/8 -too low?

5) Oxides in ODS steels greatly improve swelling behavior of these materials. The oxides act as sinks for vacancies and interstials. The also act as barriers to dislocation motion, further stabilizing the material at high ~~[illegible]~~ irradiation and temperatures. Oxides can ~~also [illegible]~~ also ~~experience embrittlement at high~~ ~~These materials are known for their high temp and~~

4/6

6) Ni-based Alloys have good resistance to irradiation, ~~[illegible]~~ but do have changes in hardness with temperature. Although precipitates help impede dislocation motion in these alloys, at lower temps will dissolve. At higher temps, however, the precipitates will reform. This results in an increase in hardness with T, but a decrease in ductility. ~~[illegible]~~ This helps with high temperature creep behavior as well.

ok..

5/8

-Corrosion resistance , He embrittlement ...

(7) U-Al_x fuels typically are U-Al_2, U-Al_3, and U-Al_4 thus the fissile density is higher in U-Si fuels where you have U_3Si and U_3Si_2 (higher concentration of fissile material). U_3Si does undergo swelling however due to the lower higher viscosity and higher diffusion, paired with more free volume in the lattice. This allows more space for swelling to occur via bubble formation and vacancy migration to voids. U_3Si_2 is very desirable however because the viscosity is lower ~~higher~~ higher, diffusion is lower, and there are less free volume spaces to accomodate swelling, and less ~~[illegible]~~ movement of atoms overall. So although, U_3Si has ~~more fissile material than U_3~~ a greater ratio of U to Si atoms, its swelling behavior is much less desirable for reactors.

11/12

- low swelly in UAlx

(8) ~~The ~~[illegible]~~ gamma phase in U-Mo helps stabilize the BCC phase~~
The Aluminum helps maintain the gamma phase of the material. This aluminum is the material of the surrounding plate in the fuel. This is due to the

-nope...

1/7

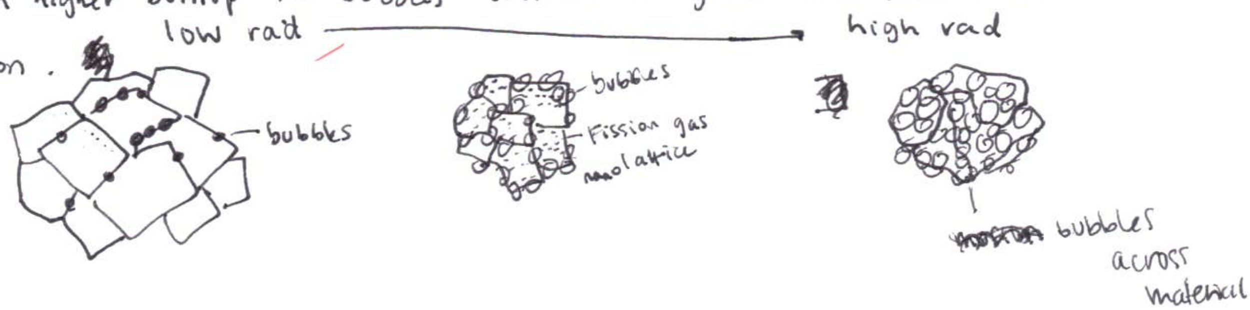
Out of time, thanks for a good semester.

⑨ U-Mo is placed into the plate in its liquid for where it bonds to the surrounding Al upon cooling. (Honestly I can't remember so this answer is probably wrong).

1/6

~yes, this is ...

⑩ At low burnup, bubbles form at Grain boundaries but not seen within the base material. As burnup increases, bubbles continue to grow along the grain boundaries and increase in size. Additionally, the fission gas nano lattice is seen (under electron microscopy) within the material. Polygimization of the nano lattice occurs and the grains decrease in size. With even higher burnup the bubbles continue to grow and form across the material cross-section.



10/10

⑪ The interaction layer in U-Mo, in dispered matrix fuel type, is a layer (of a different/new material) that forms along the outer edge of the fuel meat. This layer has a lower density than the fuel itself which leads to bubbles wanting to form at that region. This results in degredation of the material once it swells, which slowly degrades the surrounding material as well. Addition of other elements (like Si) can help slow the formation of the layer. Additionally using a monolithic fuel type with a thin Zr layer lining the outside can decrease the formation of an interaction layer as well.

7/7

⑫ Research reactors operate at lower temperatures than Fast reactors and are mostly focused on isotope production rather than power or breeding. Thus Al can better withstand these "less severe" reactor environments.

~needed more

4/6